

MATM20 Mathematical Modelling

Project Report

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In this project report, we are going to solve the Optimisation project 1: Textile distribution. Here, we are advising the Burningham textile company on their shipping needs between their textile mills and their distribution centres. Through the report, we will work through exercises:

- 15: Textile shipping
 - 21: Updating trucks
 - 22: Utilising different types of trucks
- from section 3.5 of Chapter 3 in [1]

Question 15 The Burningham textiles company has three textile mills located in the southern U.S. and four distribution centers located in Michigan, New York, California and Georgia. The estimated yearly output from each mill, the allocation to each warehouse, and the shipping costs are tabulated in the following table:

Table 1.

Mills - Locations	MI	NY	CA	GA	output
1	430	550	680	700	105
2	510	590	890	685	160
3	395	425	910	450	85
Allocation	70	100	105	75	

15(a) In this question, we are asked to find the transportation plan that minimizes shipping costs using the five-step method and to solve it as a linear programming problem using a computer.

The mathematical modeling approach to problem solving consists of five steps:

1. Ask the question.
2. Select the modeling approach.

3. Formulate the model.
4. Solve the model.
5. Answer the question.

Step 1:

Before we can ask a question in mathematical terms we go through the problem and make a list of variables, assumptions and constraints.

Variables:

x_{11} = amount shipped from textile mill 1 to Michigan
 x_{12} = amount shipped from textile mill 1 to NY
 x_{13} = amount shipped from textile mill 1 to California
 x_{14} = amount shipped from textile mill 1 to Georgia
 x_{21} = amount shipped from textile mill 2 to Michigan
 x_{22} = amount shipped from textile mill 2 to NY
 x_{23} = amount shipped from textile mill 2 to California
 x_{24} = amount shipped from textile mill 2 to Georgia
 x_{31} = amount shipped from textile mill 3 to Michigan
 x_{32} = amount shipped from textile mill 3 to NY
 x_{33} = amount shipped from textile mill 3 to California
 x_{34} = amount shipped from textile mill 3 to Georgia
 y = shipping costs (\$/year)

Assumptions and Constraints:

$b_1 = x_{11} + x_{12} + x_{13} + x_{14}$
 $b_2 = x_{21} + x_{22} + x_{23} + x_{24}$
 $b_3 = x_{31} + x_{32} + x_{33} + x_{34}$
 $a_1 = x_{11} + x_{21} + x_{31}$
 $a_2 = x_{12} + x_{22} + x_{32}$
 $a_3 = x_{13} + x_{23} + x_{33}$
 $a_4 = x_{14} + x_{24} + x_{34}$
 $y = 430x_{11} + 550x_{12} + 680x_{13} + 700x_{14} + 510x_{21} + 590x_{22} + 890x_{23} + 685x_{24} + 395x_{31} + 425x_{32} + 910x_{33} + 450x_{34}$
 $b_1 \leq 105$
 $b_2 \leq 160$
 $b_3 \leq 85$
 $a_1 \geq 70$
 $a_2 \geq 100$
 $a_3 \geq 105$
 $a_4 \geq 75$
 $x_{ij} \geq 0$, the quantity shipped from mill $i = 1, 2, 3$ to center $j = 1, 2, 3, 4$.

Where $j=1,2,3,4$ corresponds to Michigan, NY, Ca and Ga respectively.

Objective:

Minimize y , the shipping costs.

Step 2:

In this step, we will select the modeling approach. We will model this problem as a linear programming problem.

Step 3:

This step is to formulate the linear programming model in the standard form. In our problem the decision variables are $x_{11}, \dots, x_{34}$. We want to minimize the objective function $y = f(x_{11}, \dots, x_{34})$ over the feasible region defined by the constraints.

$$\begin{aligned} 1x_{11} + 1x_{12} + 1x_{13} + 1x_{14} &\leq 105 \\ 1x_{21} + 1x_{22} + 1x_{23} + 1x_{24} &\leq 160 \\ 1x_{31} + 1x_{32} + 1x_{33} + 1x_{34} &\leq 85 \\ 1x_{11} + 1x_{21} + 1x_{31} &\geq 70 \\ 1x_{12} + 1x_{22} + 1x_{32} &\geq 100 \\ 1x_{13} + 1x_{23} + 1x_{33} &\geq 105 \\ 1x_{14} + 1x_{24} + 1x_{34} &\geq 75 \end{aligned}$$

Step 4:

This step is to solve the model. The task is to minimize the total shipping cost y over the feasible region defined by the above constraints. We have used both, Python and Excel spreadsheet to compute the optimal solution. The full python code can be found under Appendix A and the Excel Table and Solver can be found in the Appendix as Fig.1 and Fig.2.

We defined a large matrix for the system. As we have only the function A_{ub} in Python taking values less than equal, we wrote the values for greater than equal as -1, and as well as the values of the constraints for that part have the minus symbol. For clarity we have shown it.

In Python the matrix can be written:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}$$

and the constraint can be written as a vector in the following way:

$$\mathbf{v} = (105, 160, 85, -70, -100, -105, -75)^t.$$

The optimal solution is $y = 198200$, at $x_{11} = 0, x_{12} = 0, x_{13} = 105, x_{14} = 0, x_{21} = 70, x_{22} = 90, x_{23} = 0, x_{24} = 0, x_{31} = 0, x_{32} = 10, x_{33} = 0, x_{34} = 75$. Since the slack variables are zero, the constraints are binding.

Step 5:

This step is to answer the question posed originally in step 1, i.e. to find a transportation plan which minimizes the shipping costs. The optimal solution is reported in form of a table:

Table 2.

Mills - Locations	MI	NY	CA	GA
1	0	0	105	0
2	70	90	0	0
3	0	10	0	75

Hence, the optimal shipping plan is to ship all of the output of mill 1 (105 truckloads) to the California distribution center. From mill 2 we will ship 70 truckloads to Michigan and 90 truckloads to New York. From mill 3 we will ship 10 truckloads to New York and 75 truckloads to Georgia. This plan will result in the lowest possible shipping cost for our current facilities. The estimated shipping costs is \$198,200 per year.

15(b) In this question, we need to determine the shadow prices for each of the output constraints and find out if it would be beneficial to shift production capacity from one mill to another. We have to also find out how much the company should be willing to spend to facilitate the shift.

Using the code and values found in **15(a)**, we can find the shadow prices for each of the output constraints. The corresponding Excel report and Solver can be found in the Appendix as Fig.2, Fig.3, and Fig.4.

The first three constraints represent the output from each mill. The shadow prices for mills 1,2,3 are \$80, \$0 and \$165 respectively. If we shift production capacity from mill 2 to mill 1 it will save the company \$80 per truckload per year, and if we shift production capacity from mill 2 to mill 3 it will save the company \$165 per truckload per year. This leads to the fact that the company should be willing to spend \$245 per truckload per year to facilitate the shift.

Note: The shadow prices differ from the Lagrange multipliers, the shadow price for mill 1 is 80, for mill 2 is 0 and for mill 3 is 165. While using Lagrange multipliers are 209.99, for mill 1, for mill 2 is 0 and for mill 3 is 165. If we look at the allowable decrease for these values, it's zero, which leads

to the conclusion that our primal solution is degenerate, and our dual problem has multiple optima. That's the reason why if we use simplex solver or GRG solver in Excel we have different values. [2], [3] If instead of using Excel, we use Python with `linprog(method='highs-ipm')`, we will have a shadow price of 210 for mill 1, using `linprog(method='highs-ds')` will return us a value equals to 80. Using `linprog(method='highs')`, it chooses between the two automatically, giving us 80, as `linprog(method='highs-ds')` does. The reason behind that is because `linprog(method='highs-ds')` is designed for dual-simplex, while `linprog(method='highs-ipm')` is for interior point method.

Question 21 This is a variation of **Question 15**. The Burningham textiles company has three textile mills located in the southern U.S. and four distribution centers located in Michigan, New York, California, and Georgia. The estimated yearly output from each mill, the allocation to each warehouse, and the shipping costs are tabulated in Table 3.

21(a) In this question we are asked to find the transportation plan that minimizes shipping costs using the five-step method and to solve it as a linear programming problem using a computer.

The solution is the same as **15(a)** above.

21(b) In this question, the company is said to have purchased three new trucks for shipping. The new trucks are more fuel-efficient, and are expected to reduce shipping costs by 50%. We are asked to assume that one truck can make one trip per week (52 per year), and that each route will use either all old or all new trucks. We need to find out how the company should use the new trucks in order to maximize their cost savings. We need to do this using the five-step method, and solve as an integer programming problem using a computer.

Table 3 summarises the values obtained in the previous exercise. As it's a binary problem we will define with $i \in \{1, 2, 3\}$ the mills, and as $j \in \{A, B, C, D\}$ the locations corresponding respectively to Michigan, New York, California and Georgia. We want to ship from site i to site j .

Table 3. Allocation

Mills - Locations	A	B	C	D
1	0	0	105	0
2	70	90	0	0
3	0	10	0	75

The new trucks are expected to reduce the shipping cost by 50%, so, we will find the new shipping costs, by dividing the previous shipping costs by 2. In Table 3 we see the distribution of the products that the trucks need to transport from the mills to the locations, we can multiply it by the new shipping costs to actually know how much the company can save.

The optimal solution found in **15(a)** is to ship on 5 routes as shown in Table 4 below. The table also shows the calculated new shipping costs and the savings.

Table 4. Routes

Routes	Allocation	Previous Cost	New Cost	Cost Savings	Trucks
1A	0	430	215	$0 * 215 = 0$	0
1B	0	550	275	$0 * 275 = 0$	0
1C	105	680	340	$105 * 340 = 35700$	3
1D	0	700	350	$0 * 350 = 0$	0
2A	70	510	255	$70 * 255 = 17850$	2
2B	90	590	295	$90 * 295 = 26550$	2
2C	0	890	445	$0 * 445 = 0$	0
2D	0	685	342,5	$0 * 342,5 = 0$	0
3A	0	395	197,5	$0 * 197,5 = 0$	0
3B	10	425	212,5	$10 * 212,5 = 2125$	1
3C	0	910	455	$0 * 455 = 0$	0
3D	75	450	225	$75 * 225 = 16875$	2

We assume that each new truck can make 52 trips per year by that we know that we will need 3 trucks for the allocation from mill 1 to location C, as we can divide 105 in 3 parts ($52 + 52 + 1$), we need 2 trucks for the distribution from mill 2 to location A, as we can divide 70 in two parts ($52 + 18$), we need 2 trucks to move products from mill 2 to location B ($90 = 52 + 38$), we need just 1 truck from mill 3 to location B ($10 < 52$), and we need 2 trucks from 3 to D ($75 = 52 + 23$). These calculations of the number trucks is listed down in the above table.

Using the five-step method:

Step 1:

Before we can ask a question in mathematical terms we go through the problem and make a list of variables, assumptions and constraints.

Variables:

$x_{i,j}$ = trucks used from site i to site j .

$p_{i,j}$ = previous cost of shipping from site i to site j (\$)

$s_{i,j}$ = cost of shipping reduced from site i to site j (\$)

S = total cost savings (\$)
 T = total number of new trucks used.

Assumptions and Constraints:

$$\begin{aligned} T &= 3x_{1C} + 2x_{2A} + 2x_{2B} + 1x_{3B} + 2x_{3D} \\ S &= 35700x_{1C} + 17850x_{2A} + 26550x_{2B} + 2125x_{3B} + 16875x_{3D} \\ T &\leq 3 \\ x_{i,j} &\text{ is a binary (0,1) integer} \end{aligned}$$

Objective:

Maximize S

Step 2:

In this step, we will select the modeling approach. We will model this problem as a binary integer programming problem using a spreadsheet.

Step 3:

This step is to formulate the binary integer programming model. In this problem the decision variables are the number of trucks $x_{i,j}$ used to ship for sites $i = 1, 2, 3$ to $j = A, B, C, D$ and our objective is to maximize the total savings $y = S$, where:

$$y = 35700x_{1C} + 17850x_{2A} + 26550x_{2B} + 2125x_{3B} + 16875x_{3D}$$

subject to the constraint:

$$3x_{1C} + 2x_{2A} + 2x_{2B} + 1x_{3B} + 2x_{3D} \leq 3$$

where the values $x_{i,j}$ are binary, they can take values 0 or 1 if the new trucks are assigned or 0 if they are not assigned.

Step 4:

This step is to solve the model. The task is to maximise the total cost savings y over the feasible region defined by the above constraints. We use Microsoft Excel to compute the optimal solution. Especially, the solver tool has been used to find the optimal solution. The Excel tables used to solve this problem can be found in the Appendix as Fig.5 and Fig.6.

The optimal solution is $y = 35700$ at $x_{1C} = 1, x_{2A} = 0, x_{2B} = 0, x_{3B} = 0, x_{3D} = 0$.

Step 5:

This step is to answer the question posed originally in step 1, i.e. to find out how to use the new trucks to maximise the total savings.

The optimal solution is to use all the three new trucks from mill 1 to location C, as for each route we can use all the new trucks or all the old trucks. This should yield the expected total cost savings for this transportation plan, which is, \$35,700 per year.

21(c) In this question, we need to perform a sensitivity analysis on the number of new trucks and determine the optimal shipping plan and the expected cost savings for $n = 4, 5, 6$, and 7 new trucks. We need to also find out that if the cost of a new truck is \$12,000 per year amortized over the expected life of the vehicle, how many additional trucks should the company purchase.

We repeat the steps in part **21(b)** but vary the constraint $T \leq n$ where $n = 3$ are the new trucks solved in **21(b)**.

For $n = 4$:

$$3x_{1C} + 2x_{2A} + 2x_{2B} + 1x_{3B} + 2x_{3D} \leq 4.$$

In step 4 (solving the model), we use the Excel spreadsheet to find the optimal solution to be 44400.

In Step 5 (answering the question), the optimal solution is to employee the new four trucks in routes from mill 2 to destination A, and from mill 2 to destination B. As each of these routes need two trucks. This should yield \$44400.

For $n = 5$:

$$3x_{1C} + 2x_{2A} + 2x_{2B} + 1x_{3B} + 2x_{3D} \leq 5.$$

In step 4 (solving the model), we use the Excel spreadsheet to find the optimal solution to be 62250.

In Step 5 (answering the question), the optimal solution is to employee the new five trucks in routes from mill 1 to destination C, and from mill 2 to destination B. As each of these routes need three and two trucks respectively. This should yield \$62250.

For $n = 6$:

$$3x_{1C} + 2x_{2A} + 2x_{2B} + 1x_{3B} + 2x_{3D} \leq 6.$$

In step 4 (solving the model), we use the Excel spreadsheet to find the optimal solution to be 64375.

In Step 5 (answering the question), the optimal solution is to employee the new six trucks in routes from mill 1 to destination C, from mill 2 to destination B, from mill 3 to destination B. As each of these routes need three, two and one truck respectively. This should yield \$64375.

For $n = 7$:

$$3x_{1C} + 2x_{2A} + 2x_{2B} + 1x_{3B} + 2x_{3D} \leq 7.$$

In step 4 (solving the model), we use the Excel spreadsheet to find the optimal solution to be 80100.

In Step 5 (answering the question), the optimal solution is to employee the new seven trucks in routes from mill 1 to destination C, from mill 2 to destination A, from mill 2 to destination B. As each of these routes need three, two and two trucks respectively. This should yield \$80100.

The results are tabulated below:

Table 5. Trucks

Routes	Allocation	Past Cost	New Cost	Cost Savings	n = 4	n = 5	n = 6	n = 7
1A	0	430	215	0	0	0	0	0
1B	0	550	275	0	0	0	0	0
1C	105	680	340	35700	0	3	3	3
1D	0	700	350	0	0	0	0	0
2A	70	510	255	17850	2	0	0	2
2B	90	590	295	26550	2	2	2	2
2C	0	890	445	0	0	0	0	0
2D	0	685	342,5	0	0	0	0	0
3A	0	395	197,5	0	0	0	0	0
3B	10	425	212,5	2125	0	0	1	0
3C	0	910	455	0	0	0	0	0
3D	75	450	225	16875	0	0	0	0

Each truck costs \$12,000 per year, the use of new trucks is justified only if they save the company more money than they cost. In table 6 the costs and savings are reported. The optimal decision is for the company to purchase 2 additional new trucks, for a total of 5 new trucks. This should result in a small net savings of $62250 - 60000 = \$2,250$ per year.

Table 6. Costs and savings

Trucks	4	5	6	7
Costs	48000	60000	72000	84000
Savings	44400	62250	64375	80100

21(d) In this question it has been mentioned that because the State of California has instituted new pollution laws for trucks, the company must use the new trucks to ship to California. We need to find out how this changes the results of parts **21(b)** and **21(c)** and what the new California pollution laws would cost

the company.

We repeat parts **21(b)** and **21(c)** but add the constraint $x_{1C} = 1$, this is because the company can use just new trucks for the route to California. The results are exactly the same as before in cases where $x_{1C} = 1$ is optimal. The only change is for $n = 4$ new trucks. If the company buys only 3 new trucks, or if they purchase the 2 additional new trucks for a total of 5 new trucks, then the California pollution law costs the company nothing.

Table 7. Trucks

Routes	Allocation	New Cost	Cost Savings	n=3	n=4	n=5	n=6	n=7
1A	0	215	0	0	0	0	0	0
1B	0	275	0	0	0	0	0	0
1C	105	340	35700	3	3	3	3	3
1D	0	350	0	0	0	0	0	0
2A	70	255	17850	0	0	0	0	2
2B	90	295	26550	0	0	2	2	2
2C	0	445	0	0	0	0	0	0
2D	0	342,5	0	0	0	0	0	0
3A	0	197,5	0	0	0	0	0	0
3B	10	212,5	2125	0	1	0	1	0
3C	0	455	0	0	0	0	0	0
3D	75	225	16875	0	0	0	0	0

Table 8. Costs and savings

Trucks	3	4	5	6	7
Costs	36000	48000	60000	72000	84000
Savings	35700	37825	62250	64375	80100

Thus, the company should buy 5 new trucks. The Excel Table for this can be found in the Appendix as Fig.7.

Question 22 Here, we are asked to reconsider the textile shipping problem of **Question 21**, but now we need to assume that both old and new trucks can be used on the same route. Then we have to answer the same set of questions **21(a)** to **21(d)** as in **Question 21**. [Hint: Use one decision variable to indicate whether or not one new truck is used on route i , another decision variable to

indicate whether a second truck is used, and so forth.]

22(a) In this question we are asked to find the transportation plan that minimizes shipping costs using the five-step method and to solve it as a linear programming problem using a computer.

The solution is the same as **15(a)** above.

The optimal solution is to ship on the routes as shown in table 4. We have also tabulated the potential cost savings per route and the number of trucks required to service each route in the table 4.

22(b) In this question, the company is said to have purchased three new trucks for shipping. The new trucks are more fuel-efficient, and are expected to reduce shipping costs by 50%. We are asked to assume that one truck can make one trip per week (52 per year), and that each route will use either all old or all new trucks. We need to find out how the company should use all the trucks in order to maximize their cost savings. We need to do this using the five-step method, and solve as an integer programming problem using a computer.

We assume that each new truck can make 52 trips per year. By that we know that we will need 3 trucks for the allocation from mill 1 to location C, as we can divide 105 in 3 part, we need 2 trucks for the allocation from mill 2 to location A, as we can divide 70 in two parts, we need 2 trucks to move products from mill 2 to location B, we need just 1 truck from mill 3 to location B, and we need 2 trucks from 3 to D.

The following equations report examples of division which can be made:

$$105 = 52 + 52 + 1$$

$$70 = 52 + 18$$

$$90 = 52 + 38$$

$$75 = 52 + 23$$

The reader should notice that from mill 3 to location B we need just one truck, as $10 < 52$. After trying different possibilities, like $70 = 35 + 35$, $90 = 45 + 45$, we ended up to the conclusion that assigning routes to the trucks until their limit capacity (52) is a better option, otherwise the company will save less money.

The following table summarizes the computations that we have done:

Table 9. Routes

Routes	Variables	Allocation	Previous Cost	New Cost	Cost Savings
1C	x_1	52	680	340	$52 * 340 = 17680$
1C	x_2	52	680	340	$52 * 340 = 17680$
1C	x_3	1	680	340	$1 * 340 = 340$
2A	x_4	52	510	255	$52 * 255 = 13260$
2A	x_5	18	510	255	$18 * 255 = 4590$
2B	x_6	52	590	295	$52 * 295 = 15340$
2B	x_7	38	590	295	$38 * 295 = 11210$
3B	x_8	10	425	212,5	$10 * 212,5 = 2125$
3D	x_9	52	450	225	$52 * 225 = 11700$
3D	x_{10}	23	450	225	$23 * 225 = 5175$

Using the five-step method:

Step 1:

Before we can ask a question in mathematical terms we go through the problem and make a list of variables, assumptions and constraints.

Variables:

x_i = trucks used as shown in table 9 where $i = 1, 2, \dots, 10$.

$p_{i,j}$ = previous cost of shipping from site i to site j (\$)

$s_{i,j}$ = cost of shipping reduced from site i to site j (\$)

S = total cost savings (\$)

T = total number of new trucks used.

Assumptions and Constraints:

$$T = x_1 + x_2 + \dots + x_{10}$$

$$S = 17680x_1 + 17680x_2 + 340x_3 + 13260x_4 + 4590x_5 + 15340x_6 + 11210x_7 + 2125x_8 + 11700x_9 + 5175x_{10}$$

$$T \leq 3$$

x_i is a binary (0,1) integer

Objective:

Maximize S

Step 2:

In this step, we will select the modeling approach. We will model this problem as a binary integer programming problem using a spreadsheet.

Step 3:

This step is to formulate the binary integer programming model. In this problem the decision variables are the trucks x_i used to ship and our objective is to maximize the total savings $y = S$, where:

$$y = 17680x_1 + 17680x_2 + 340x_3 + 13260x_4 + 4590x_5 + 15340x_6 + 11210x_7 + 2125x_8 + 11700x_9 + 5175x_{10}$$

subject to the constraint:

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} \leq 3$$

where the values x_i are binary for $i = 1, 2, \dots, 10$, they can take values 0 or 1, 1 if the new trucks are assigned or 0 if they are not assigned.

Step 4:

This step is to solve the model. The task is to maximise the total cost savings y over the feasible region defined by the above constraints. We use Microsoft Excel to compute the optimal solution. Especially, the solver tool has been used to find the optimal solution. The Excel table used to solve this problem can be found in the Appendix as Fig.8.

The optimal solution is $y = 50700$ at $x_1 = 1, x_2 = 1, x_6 = 1$ and the remaining $x_i = 0$.

Step 5:

This step is to answer the question posed originally in step 1, i.e. to find out how to use all the trucks to maximise the total savings.

The optimal solution is to use the new trucks on route assignments x_1, x_2 (52 trips per year on route 1 from mill 1 to location C) and x_6 (52 trips per year on route 3 from mill 2 to location B). This will use all 3 new trucks on 52 trips per year. The expected total cost savings of this transportation plan is \$50700 per year.

22(c) In this question, we need to perform a sensitivity analysis on the number of new trucks and determine the optimal shipping plan and the expected cost savings for $n = 4, 5, 6$, and 7 new trucks. We need to also find out that if the cost of a new truck is \$12,000 per year amortized over the expected life of the vehicle, how many additional trucks should the company purchase.

We repeat part **22(b)**. Using the five-step method:

Step 1:

Before we can ask a question in mathematical terms we go through the problem and make a list of variables, assumptions and constraints.

Variables:

x_i = trucks used as shown in table 9 where $i = 1, 2, \dots, 10$.

$p_{i,j}$ = previous cost of shipping from site i to site j (\$)

$s_{i,j}$ = cost of shipping reduced from site i to site j (\$)

S = total cost savings (\$)

T = total number of new trucks used.

Assumptions and Constraints:

$$T = x_1 + x_2 + \dots + x_{10}$$

$$S = 17680x_1 + 17680x_2 + 340x_3 + 13260x_4 + 4590x_5 + 15340x_6 + 11210x_7 + 2125x_8 + 11700x_9 + 5175x_{10}$$

$$T \leq n \text{ where } n = 4, 5, 6, 7$$

x_i is a binary (0,1) integer

Objective:

Maximize S

Step 2:

In this step, we will select the modeling approach. We will model this problem as a binary integer programming problem using a spreadsheet.

Step 3:

This step is to formulate the binary integer programming model. In this problem the decision variables are the trucks x_i used to ship and our objective is to maximize the total savings $y = S$, where:

$$y = 17680x_1 + 17680x_2 + 340x_3 + 13260x_4 + 4590x_5 + 15340x_6 + 11210x_7 + 2125x_8 + 11700x_9 + 5175x_{10}$$

subject to the constraint:

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} \leq n$$

where $n = 4, 5, 6, 7$ and the values x_i are binary for $i = 1, 2, \dots, 10$, they can take values 0 or 1 if the new trucks are assigned or 0 if they are not assigned.

Step 4:

This step is to solve the model. The task is to maximise the total cost savings y over the feasible region defined by the above constraints. We use Microsoft Excel to compute the optimal solution. Especially, the solver tool has been used to find the optimal solution. The Excel table and solver used to solve this problem can be found in the Appendix as Fig.8 and Fig.9. The optimal solutions are tabulated in Table 10 and Table 11.

Step 5:

This step is to answer the question posed originally in step 1, i.e. to find out how to use all the trucks to maximise the total savings.

The optimal solution is that the company should purchase 4 new trucks in total, as we will have the maximum saving equals to \$15,960.

Table 10. Routes

Routes	n = 4	n = 5	n = 6	n = 7
1C	1	1	1	1
1C	1	1	1	1
1C	0	0	0	0
2A	1	1	1	1
2A	0	0	0	0
2B	1	1	1	1
2B	0	0	1	1
3B	0	0	0	0
3D	0	1	1	1
3D	0	0	0	1

Table 11. Trucks and Savings

trucks	3	4	5	6	7
costs	36000	48000	60000	72000	84000
savings	50700	63960	75660	86870	92045
shadow prices	13260	11700	11210	5175	4590

22(d) In this question it has been mentioned that because the State of California has instituted new pollution laws for trucks, the company must use the new trucks to ship to California. We need to find out how this changes the results of parts **22(b)** and **22(c)** and what the new California pollution laws would cost the company.

We repeat the same process of parts **22(b)** and **22(c)**, but since the truck must be new ones for the route to California because of the new laws, we don't have to use binary in Excel solver for this particular route, but we will put the values of this route equal to 1. That is, we add the constraints of variables x_1, x_2, x_3 from Table 9 as $x_1 = 1, x_2 = 1$, and $x_3 = 1$.

We have shown our results in the Table 12 and Table 13. The Excel table and solver used to solve this problem can be found in the Appendix as Fig.8 and Fig.10.

Table 12. Routes

Routes	n = 3	n = 4	n = 5	n = 6	n = 7
1C	1	1	1	1	1
1C	1	1	1	1	1
1C	1	1	1	1	1
2A	0	0	1	1	1
2A	0	0	0	0	0
2B	0	1	1	1	1
2B	0	0	0	0	1
3B	0	0	0	0	0
3D	0	0	0	1	1
3D	0	0	0	0	0

Table 13. trucks and cost saving

Trucks	3	4	5	6	7
costs	36000	48000	60000	72000	84000
savings	35700	51040	64300	76000	87210
savings - cost	-300	3040	4300	4000	3210
shadow prices	15340	13260	11700	11210	5175

The company should purchase 5 trucks as the maximum savings will be obtained. The net savings is \$4,300 per year. The CA pollution law costs the company an amount equals to \$11,360 per year for this transportation plan. This value is obtained by taking the difference between the previous value of total savings for $n=5$ obtained in part **22(c)**, i.e \$75,660 and the current value after the new law \$64,300.

We notice that the shadow price for 3 trucks is different from the Lagrange multiplier, the last resulting in 5175. Looking at the allowable decrease we encounter a value equal to zero. This means that our primal solution is degenerate and we have multiple optimal solutions. So the two solvers found different dual solutions. [3]

A Appendix

15(a): Python Code:

```

1 import numpy as np
2 from scipy.optimize import linprog
3
4 A_ub = np.array([
5     [1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0],
6     [0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0],
7     [0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1],
8     [-1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0],
9     [0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0, 0],
10    [0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1, 0],
11    [0, 0, 0, -1, 0, 0, 0, -1, 0, 0, 0, -1]
12])
13 print(A_ub)
14
15 b_ub = np.array([105, 160, 85, -70, -100, -105, -75])
16 print(b_ub)
17
18 c = np.array([430, 550, 680, 700, 510, 590, 890, 685, 395,
19               425, 910, 450])
20
21 res = linprog(c, A_ub=A_ub, b_ub=b_ub, bounds=(0, None),
22               method='highs')
23
24 print(res)
25
26 y = res.fun
27 print('Shipping costs: {:.1f}'.format(y))
28
29 for idx, val in enumerate(res.x):
30     print(f" {val:.6f} ")
31
32 slack = res.slack
33 print(f" {slack[0]:.6f} ")
34 print(f" {slack[1]:.6f} ")
35 print(f" {slack[2]:.6f} ")
36
37 res.ineqlin
38
39 shadow_pr = -res.ineqlin.marginals
40 shadow_pr

```

Note for the reader: For exercises 21(c),21(d), the table is the same as 21(b), only the solver is changing with the constraints, we uploaded the images of just some cases of it, but it's possible to read all the results during the report. This is valid also for exercise 22.

Excel Tables and Solvers for Q15,21,22:

	A	B	C
1		variables	allocation
2	1	x11	0
3	2	x12	0
4	3	x13	105
5	4	x14	0
6	5	x15	70
7	6	x16	90
8	7	x17	0
9	8	x18	0
10	9	x19	0
11	10	x20	10
12	11	x21	0
13	12	x22	75
14		objective function	=SUM(430*C2+550*C3+680*C4+700*C5+510*C6+590*C7+890*C8+685*C9+395*C10+425*C11+910*C12+450*C13)
15	constraints		
16	=SUM(C2+C3+C4+C5)	<=	105
17	=SUM(C6+C7+C8+C9)	<=	160
18	=SUM(C10+C11+C12+C13)	<=	85
19	=SUM(C2+C6+C10)	>=	70
20	=SUM(C3+C7+C11)	>=	100
21	=SUM(C4+C8+C12)	>=	105
22	=SUM(C5+C9+C13)	>=	75
23			

Fig. 1. 15(a): Excel Table

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

- \$A\$16 <= 105
- \$A\$17 <= 160
- \$A\$18 <= 85
- \$A\$19 >= 70
- \$A\$20 >= 100
- \$A\$21 >= 105
- \$A\$22 >= 75

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for Linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Fig. 2. 15: Excel Solver

Variable Cells						
Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$C\$1	x11	0	0	430	40	130
\$C\$2	x12	0	40	550	1E+30	40
\$C\$3	x13	105	0	680	130	760
\$C\$4	x14	0	165	700	1E+30	165
\$C\$5	x15	70	0	510	50	40
\$C\$6	x16	90	0	590	40	50
\$C\$7	x17	0	130	890	1E+30	130
\$C\$8	x18	0	70	685	1E+30	70
\$C\$9	x19	0	50	395	1E+30	50
\$C\$10	x20	10	0	425	50	70
\$C\$11	x21	0	315	910	1E+30	315
\$C\$12	x22	75	0	450	70	615

Constraints						
Cell	Name	Final Value	Shadow Price	Constraint R.H.Side	Allowable Increase	Allowable Decrease
\$A\$15		105	-80	105	70	0
\$A\$16		160	0	160	1E+30	0
\$A\$17		85	-165	85	90	0
\$A\$18		70	510	70	0	70
\$A\$19		100	590	100	0	90
\$A\$20		105	760	105	0	70
\$A\$21		75	615	75	0	75

Fig. 3. 15(b): Shadow Prices

Variable Cells			
Cell	Name	Final Value	Reduced Gradient
\$C\$1	x11	0	129,9999827
\$C\$2	x12	0	169,9999676
\$C\$3	x13	105	0
\$C\$4	x14	0	294,9999431
\$C\$5	x15	70	0
\$C\$6	x16	90	0
\$C\$7	x17	0	0
\$C\$8	x18	0	69,99998789
\$C\$9	x19	0	50,00000851
\$C\$10	x20	10	0
\$C\$11	x21	0	185,0000451
\$C\$12	x22	75	0

Constraints			
Cell	Name	Final Value	Lagrange Multiplier
\$A\$15		105	-209,9999462
\$A\$16		160	0
\$A\$17		85	-165,0000098
\$A\$18		70	510,0000118
\$A\$19		100	590,0000045
\$A\$20		105	889,999962
\$A\$21		75	615,0000107

Fig. 4. 15(b): Lagrange Multipliers

	A	B	C	D	E	F	G
1		variables	allocation	shipping cost	saving	total savings	new trucks
2	1	x1	0	430	=1/2*D2	=E2*C2	0
3	2	x2	0	550	=1/2*D3	=E3*C3	0
4	3	x3	105	680	=1/2*D4	=E4*C4	1
5	4	x4	0	700	=1/2*D5	=E5*C5	0
6	5	x5	70	510	=1/2*D6	=E6*C6	0
7	6	x6	90	590	=1/2*D7	=E7*C7	0
8	7	x7	0	890	=1/2*D8	=E8*C8	0
9	8	x8	0	685	=1/2*D9	=E9*C9	0
10	9	x9	0	395	=1/2*D10	=E10*C10	0
11	10	x10	10	425	=1/2*D11	=E11*C11	0
12	11	x11	0	910	=1/2*D12	=E12*C12	0
13	12	x12	75	450	=1/2*D13	=E13*C13	0
14							
15	objective function	=SUM(430*(			objective function for max saving	=SUM(F4*G4+F6*G6+F7*G7+F11*G11+F13*G13)	
16							
17	constraints				constraint		
18	=SUM(C2+C3+C4+C5)	<=	105			=SUM(3*G4+2*G6+2*G7+G11+2*G13)	
19	=SUM(C6+C7+C8+C9)	<=	160				
20	=SUM(C10+C11+C12+C13)	<=	85				
21	=SUM(C2+C6+C10)	>=	70				
22	=SUM(C3+C7+C11)	>=	100				
23	=SUM(C4+C8+C12)	>=	105				
24	=SUM(C5+C9+C13)	>=	75				
25							

Fig. 5. 21(b): Excel Table

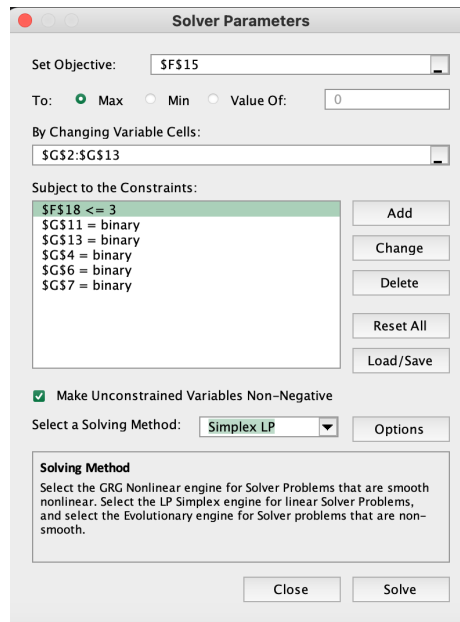


Fig. 6. 21(b): Excel Solver

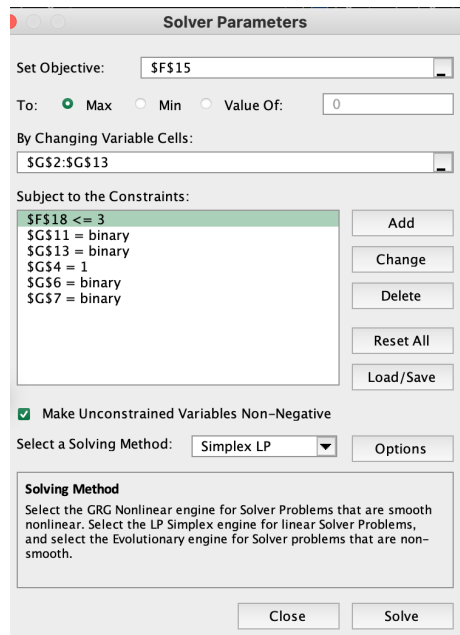


Fig. 7. 21(d): Excel Solver

	A	B	C	D	E	F	G
1		variables	allocation	route	cost	savings	total savings
2	1	x1	52	1 to Cali	680	=1/2*E2	=C2*F2
3	1	x2	52	1 to Cali	680	=1/2*E3	=C3*F3
4	1	x3	1	1 to Cali	680	=1/2*E4	=C4*F4
5	1	x4	52	2 to Mi	510	=1/2*E5	=C5*F5
6	0	x5	18	2 to Mi	510	=1/2*E6	=C6*F6
7	1	x6	52	2 to Ny	590	=1/2*E7	=C7*F7
8	1	x7	38	2 to Ny	590	=1/2*E8	=C8*F8
9	0	x8	10	3 to Ny	425	=1/2*E9	=C9*F9
10	1	x9	52	3 to Ga	450	=1/2*E10	=C10*F10
11	0	x10	23	3 to Ga	450	=1/2*E11	=C11*F11
12							
13		objective function	=SUM(G2*A2+G3*A3+G4*A4+G5*A5+G6*A6+G7*A7+G8*A8+G9*A9+G10*A10+G11*A11)				
14		constraint	relation	limit			
15		=SUM(A2:A11)	<=	7			
16							

Fig. 8. 22: Excel Table

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

Add Change Delete Reset All Load/Save

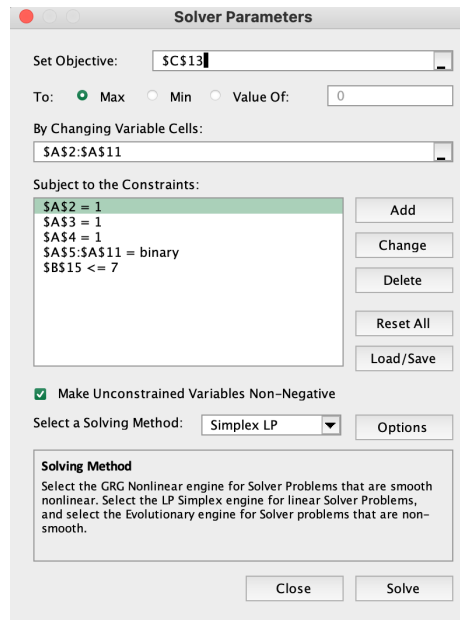
☒ Make Unconstrained Variables Non-Negative

Select a Solving Method: Options

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Close Solve

Fig. 9. 22(c): Excel Solver



The image shows the 'Solver Parameters' dialog box in Microsoft Excel. The 'Set Objective' field is set to '\$C\$13'. The 'To:' section has 'Max' selected, with 'Min' and 'Value Of:' (set to 0) as options. The 'By Changing Variable Cells' field is set to '\$A\$2:\$A\$11'. The 'Subject to the Constraints' list contains: '\$A\$2 = 1', '\$A\$3 = 1', '\$A\$4 = 1', '\$A\$5:\$A\$11 = binary', and '\$B\$15 <= 7'. To the right of this list are buttons for 'Add', 'Change', 'Delete', 'Reset All', and 'Load/Save'. Below the constraints is a checked checkbox for 'Make Unconstrained Variables Non-Negative'. The 'Select a Solving Method' dropdown is set to 'Simplex LP', with an 'Options' button next to it. A 'Solving Method' text box provides instructions on selecting the GRG Nonlinear engine for smooth nonlinear problems, the LP Simplex engine for linear problems, and the Evolutionary engine for non-smooth problems. At the bottom are 'Close' and 'Solve' buttons.

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

-
-
-
-
-

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Fig. 10. 22(d): Excel Solver

References

1. Mark M. Meerschaert, Mathematical Modeling, 4th edition, Academic press, 2013, ISBN: 9780123869128
2. Chi-Jen Lin (2010). Computing shadow prices/costs of degenerate LP problems with reduced simplex tables. Expert Systems with Applications, Volume 37, Issue 8, Pages 5848-5855, ISSN 0957-4174, <https://doi.org/10.1016/j.eswa.2010.02.021>.
3. StackExchange. 'Why is the Lagrange Multiplier not equal to the Shadow Price? (Excel Solver, MATLAB linprog, Gurobi).' URL: <https://or.stackexchange.com/questions/4946/why-is-the-lagrange-multiplier-not-equal-the-shadow-price-excel-solver-matlab>